# mmMotion: A Real-Time Human Motion Detection Smart Home System Based on mmWave Radar

Runqi Zeng, Jiuzhou Zhang, Yifan Mao, Zhaohua Yang, and Ling Shi, *Fellow, IEEE*

***Abstract*—The use of millimeter-wave radar (mmWave radar) for human motion detection is essential in the realm of smart home Internet of Things (IoT) scenarios due to its nonintrusive nature, privacy considerations, interactive capabilities, and cost-effectiveness. However, the existing methods fail to address the issue of low real-time detection rates. Moreover, the design of the action does not align with the requirements of smart home IoT systems, while the complexity of the algorithm leads to high computing hardware costs. To address these problems, we propose mmMotion, a novel mmWave radar-based human motion detection system capable of accurately estimating six well-designed human motions (sit down, stand up, get up, lie down, wave hands, and punch) commonly observed in smart home scenarios with cost-effectiveness and real-time processing speed. By incorporating the spatial feature of "height," we are the first to propose a novel feature extraction method using the Doppler–height–time (DHT) feature maps as system input for human motion detection. A combined lightweight convolutional neural network (CNN) and long short-term memory (LSTM) architecture is developed to simultaneously capture spatial and temporal features, leveraging the strengths of both architectures. Furthermore, we propose a novel real-time classifier combined with the static clutter removal (SCR) method and the Duo Filter algorithm to enhance the real-time detection capability of mmMotion. Experimental results demonstrate that mmMotion achieves an impressive offline training accuracy rate of 98.13% and a real-time motion recognition rate of 90.5% for six designated motions. Furthermore, it exhibits excellent generalization ability across different individuals and robustness to the interference of large and small actions during real-time performance testing.**

***Index Terms*—Human activity recognition, human motion detection, Internet of Things (IoT), millimeter-wave (mmWave) radar.**

## I. Introduction

### A. Background and Motivation

The ability to detect and analyze human motion is crucial for a wide range of applications, including security surveillance, healthcare monitoring, smart home technologies, indoor mapping, and human–computer interaction [1], [2], [3], [4], [5], [6], [7]. Numerous methodologies for human motion detection have been extensively investigated [8], [9], [10], [11], [12], [13]. However, conventional motion detection methods such as infrared sensors and cameras often encounter limitations in terms of precision, privacy concerns, lighting conditions, and adaptability to diverse environments [14], [15], [16]. This is where millimeter-wave (mmWave) radar technology emerges as a promising solution [17].

Millimeter-wave radar (mmWave) offers a multitude of advantages that make it highly effective in human motion detection. Notable attributes include its nonintrusive operation, compact antenna size, and ability to capture high-resolution imagery [18], [19], [20]. By harnessing the detailed data provided by mmWave radar, sophisticated algorithms can be developed for spatial positioning analysis and Doppler velocity estimation of subjects. Importantly, the noninvasive nature of mmWave radar ensures the collection of only anonymous data, thereby safeguarding individual privacy [21]. These characteristics make mmWave radar particularly suitable for addressing human motion detection tasks in smart home scenarios compared to traditional approaches such as cameras, LiDAR, and IR/Ultrasonic sensors [22], [23], [24], [25].

### B. Main Challenges

The current challenges in addressing the issue of human motion detection with mmWave radar arise from the selection of feature extraction methods and classification models, trainable motions design, as well as real-time performance and robustness of the system.

1) *Feature Extraction and Selection Methods for Classification Models:* Many conventional radar-based motion detection approaches draw inspiration from computer vision methods. Range–Doppler (RD), range–angle (RA), RD–time (RDT), and angle–Doppler–time (ADT) feature maps are commonly employed as input for the models [26]. Support vector machines (SVMs) [27], convolutional neural networks (CNNs) [28], long short-term memory (LSTM) [29], [30], and Transformers [31], [32] are frequently utilized as training and classification models.
2) *Motion Design:* The design of actions does not align with the requirements of smart home Internet of Things (IoT) systems. Many large-movement motions, such as walking and kicking, as well as small-movement motions like special gestures and shaking one's head,

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pose challenges and lack practicality within the context of a smart home scenario [33].

3) *Real-Time and Robustness:* The majority of existing research primarily focuses on offline detection as the primary research direction, thereby rendering their systems nonreal-time and less robust for intricate real-world scenarios [34].

### C. State-of-the-Art Results and Their Limitations

Additionally, researchers have explored the integration of dynamic models and filtering algorithms to stabilize skeletal pose estimation using mmWave radar. Liu et al. [27] used SVMs and $K$-nearest neighbor (KNN) to perform human posture recognition and classification. Following a similar approach, Jokanovic et al. [35] converted spectrograms representing activities such as bending, falling, sitting, and walking into monochromatic images. They subsequently utilized deep neural networks (DNNs) and SVMs for recognition tasks. However, in practical scenarios, the performance of SVMs and KNN algorithms is suboptimal [36].

Moreover, natural language processing (NLP) techniques have been employed to process mmWave radar point-cloud data and convert it into precise skeletal pose estimations [37]. This novel approach showcases the potential of Seq2Seq-based methods for achieving accurate and detailed skeletal pose estimation using mmWave radar technology. The raw data generated by mmWave radar, however, exhibits a certain level of complexity, necessitating extensive data processing and analysis to extract meaningful skeletal feature information [38]. Consequently, this complexity presents challenges in terms of algorithm intricacy and computational demands.

### D. Our Approach

To address these limitations, we propose mmMotion, a novel mmWave radar-based system that surpasses the scope of many existing studies on offline system accuracy by accurately estimating a wide range of human motions commonly observed in smart homes, processing data in real-time, and being cost-effective. Unlike previous approaches that focus solely on offline accuracy, our solution offers a comprehensive and practical approach to tracking diverse human movements within smart home environments. As depicted in Fig. 1, the proposed mmMotion system employs mmWave radar for capturing human motion, thereby serving as a trigger to regulate smart appliances in both smart bedroom and smart office scenarios. mmMotion is capable of capturing six different motions, which are sit down, stand up, get up, lie down, wave hands, and boxing.

In a bedroom scenario, the radar is positioned on the bedside table. Upon detecting a person getting up from the bed at night, the system will illuminate the entire lighting environment, eliminating the need for users to search for the light switch in the darkness. Similarly, when the radar detects a person lying down on the bed, it will automatically dim all lights and activate sleep mode. In an office setting, the radar is placed on the working table facing toward the user. When it senses someone standing up from their chair, it will deactivate the power systems of personal desk items such as lamps or decorative light sources to conserve energy. Likewise, when someone sits back down in their chair, these power systems will be reactivated. We have also incorporated two additional detectable movements: waving and punching, actions that are not commonly performed in either scenario mentioned above. Consequently, users can define these two motions to trigger switches for various smart appliances, such as controlling specific lights or operating electric curtains.

The demo video can be found at https://youtu.be/0z10HDJb2XU

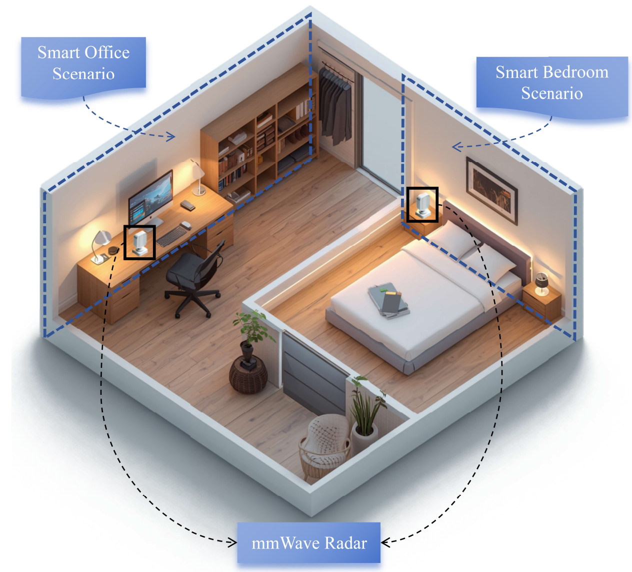



Fig. 1. Deployment of the mmWave radar on a bedside table (bedroom) and a desk (office) for motion detection.

### E. Contributions

The main contributions of this article are listed as follows.

1) In this article, we present mmMotion, which is the first work to implement real-time mmWave radar-based human motion detection and a smart appliance control system in the field of smart home scenarios. The six types of user-friendly motions have been defined for interaction in the smart home scene. mmMotion enables intelligent perception of user motions within the smart environment and intelligent control over their home environment.
2) We propose a novel feature extraction method utilizing Doppler–height–time (DHT) feature maps as system input for human motion detection using mmWave radar. Diverging from existing methods that rely on 3-D point clouds or Doppler spectra, our approach introduces a novel concept—the inclusion of “height”—to estimate human motion. We are the first to incorporate this additional spatial dimension, which sets our method apart from traditional techniques relying solely on point clouds or Doppler spectrum analysis. By introducing “height” as a key factor, our method offers a unique and pioneering approach to accurately estimating human motion.

3) We introduce an innovative approach that seamlessly integrates a lightweight CNN and LSTM. The CNN effectively captures spatial and Doppler velocity features, harnessing its power in processing spatial data. Concurrently, the LSTM excels at extracting temporal features associated with motion continuity, capitalizing on its ability to process sequential data. Our approach achieves an impressive offline training accuracy rate of 98.13%, outperforming the majority of existing studies.
4) Expanding upon our previous research [39], mmMotion presents a novel real-time classifier approach that integrates a coupled CNN–LSTM architecture and a Duo Filter, significantly augmenting its capabilities for real-time human motion detection. We have conducted an extensive experimental evaluation of the real-time classifier, with results demonstrating that our methodology achieves an impressive testing accuracy rate of 90.5%, thereby validating the effectiveness of the proposed enhancements in enabling precise real-time motion detection. Furthermore, we perform a comprehensive generalization and robustness test to evaluate mmMotion's ability to resist interference.

The subsequent sections of this article are organized as follows. Section II presents the working principle of frequency-modulated continuous-wave (FMCW) radar. Section III elucidates the mmMotion system workflow, feature extraction method, CNN–LSTM model, real-time classifier, and smart home appliances control module. Section IV showcases our experimental implementation, offline model training results, and comprehensive evaluation on different feature extraction methods, real-time classification performance, system's generalization ability and robustness, as well as system's computational efficiency and power consumption. Section V concludes this article.

## II. Preliminaries

### A. mmWave FMCW Radar Signal

The chirp is an FMCW signal that employs frequency modulation. As illustrated in Fig. 2, it operates by transmitting a sequence of frequency-modulated signals at fixed time intervals from the transmitter, with the frequency of these signals linearly varying within a predetermined range. The transmitted chirp signal $x_{\mathrm{TX}}$ can be expressed as

$$x_{\mathrm{TX}}(t) = A_{\mathrm{TX}} \cos\left[2\pi\left(f_c t + \frac{1}{2}S t^2\right) + \phi_0\right] \tag{1}$$

where $A_{\mathrm{TX}}$ is the amplitude of the transmitted chirp, $f_c$ is its center frequency, $S = B/T$ is the slope of the frequency, $\phi_0$ is the initial phase, $B$ is the bandwidth, $T$ is the duration of the chirp, and $t$ is the fast time interval $[-T/2, T/2]$.

When the transmitted chirp signal is delayed in the air with a time delay $\tau(t) = (2R - 2vt)/c$, the received chirp signal $x_{\mathrm{RX}}$ can be obtained as

$$x_{\mathrm{RX}}(t) = \mathrm{G}A_{\mathrm{TX}} \cos\left[2\pi\left(f_c(t-\tau(t)) + \frac{1}{2}S(t-\tau(t))^2\right) + \phi_0\right] \tag{2}$$

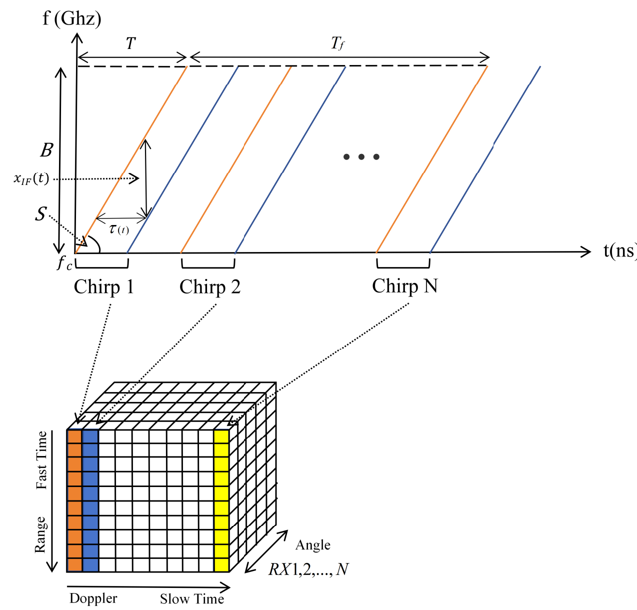



Fig. 2. Schematic representation of the FMCW chirp signal.

where $G$ is the gain multiple of the $x_{\mathrm{TX}}$ compared to $x_{\mathrm{RX}}$.

The mixed signal $x_{\mathrm{mix}}(t)$ can be derived by using trigonometric formula

$$\begin{aligned} x_{\mathrm{mix}}(t) &= x_{\mathrm{RX}}(t) \cdot x_{\mathrm{TX}}(t) \\ &= \frac{1}{2}[\underbrace{\cos(x_{\mathrm{RX}}(t) + x_{\mathrm{TX}}(t))}_{\cos(\phi_{\mathrm{sum}}(t))} + (\underbrace{\cos(x_{\mathrm{RX}}(t) - x_{\mathrm{TX}}(t))}_{\cos(\phi_{\mathrm{diff}}(t))}] \end{aligned} \tag{3}$$

After passing through the low-pass filter (LPF), the summation term $\cos(\phi_{\mathrm{sum}}(t))$ in the result will be filtered out as the high-frequency component, leaving only the low-frequency component of the difference $\cos(\phi_{\mathrm{diff}}(t))$. The intermediate-frequency (IF) signal $x_{\mathrm{IF}}(t)$ can be obtained by

$$\begin{aligned} x_{\mathrm{IF}}(t) &= \frac{1}{2}\cos(\phi_{\mathrm{diff}}(t)) = \frac{1}{2}\mathrm{G}A_{\mathrm{TX}}^2 \\ &\quad \times \cos\left[2\pi\left(f_c t - f_c\tau(t) + \frac{1}{2}S t^2 - S\tau(t)t + \frac{1}{2}S\tau(t)^2\right)\right. \\ &\quad \left. + \phi_0 - 2\pi\left(f_c t + \frac{1}{2}S t^2\right) - \phi_0\right] \\ &= \frac{1}{2}\mathrm{G}A_{\mathrm{TX}}^2 \cos\left(-2\pi f_c\tau(t) - 2\pi S\tau(t)t + \pi S\tau(t)^2\right) \\ &= A_{\mathrm{IF}} \cos\left(2\pi f_c\tau(t) + 2\pi S\tau(t)t - \pi S\tau(t)^2\right) \\ &\approx A_{\mathrm{IF}} \cos 2\pi\left(f_c\tau(t) + 2\pi S\tau(t)t\right) \end{aligned} \tag{4}$$

where $A_{\mathrm{IF}} = 1/2\mathrm{G}A_{\mathrm{TX}}^2$ is the amplitude of the IF signal. Since electromagnetic waves move at the speed of light $c$, the magnitude of time delay $\tau(t)$ for a 1-meter target ahead is roughly $10^{-8}$. The magnitude of center frequency $f_c$ is $10^9$ and the magnitude of the slope $S$ is about $10^{13}$. Comparing the magnitudes of two additional phases $2\pi f_c\tau(t) \approx 10$ and $\pi S\tau(t)^2 \approx 10^{-3}$. Therefore, the additional phase $-\pi S\tau(t)^2$ will be ignored [40].

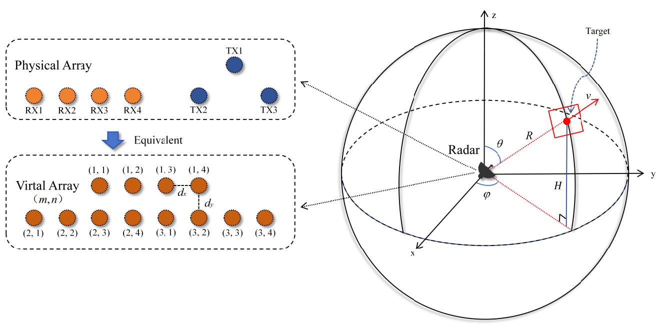



Fig. 3. Planar array geometry for mmWave radar signal processing.

Note that time delay $\tau(t) = (2R - 2vt)/c$ and $S = B/T$, the IF signal $x_{\mathrm{IF}}(t)$ can be further expressed in exponential form

$$x_{\mathrm{IF}}(t) = A_{\mathrm{IF}} e^{j2\pi\left(\overbrace{\frac{2f_c R}{c}}^{\phi_0} + \left(\overbrace{\frac{2BR}{cT}}^{f_R} - \overbrace{\frac{2f_c v}{c}}^{f_D}\right)t\right)} \tag{5}$$

where the initial phase $\phi_0 = 2f_c R/c$, the range-related frequency $f_R = 2BR/cT$, and the Doppler-related frequency shift $f_D = 2f_c v/c$.

In the radar signal-processing pipeline, fast time sampling is used to calculate the range $R$ of detected objects by analyzing the time delay of reflected signals, while slow time sampling across multiple chirps enables the estimation of Doppler velocity $v$ by observing phase variations, as shown in Fig. 2. The range $R$ and Doppler velocity $v$ is proportional to $f_R$ and $f_D$, respectively, as shown in (6a) and (6b). The range resolution $R_{\mathrm{res}}$ is determined by signal bandwidth $B$ and the maximum detection range $R_{\max}$ is determined by analog-to-digital converter (ADC) sampling frequency rate $f_S$, as given by (6c) and (6d). The Doppler velocity resolution $v_{\mathrm{res}}$ is inversely proportional to the total frame time $T_f$, as shown in (6e). The maximum detectable Doppler velocity is measured by two chirps spaced $T$ apart, as presented in (6f)

$$R = \frac{\mathrm{c}f_R T}{2B} \tag{6a}$$

$$R_{\mathrm{res}} = \frac{c}{2B} \tag{6b}$$

$$R_{\max} = \frac{c f_S T}{2B} \tag{6c}$$

$$v = \frac{c f_D}{2f_c} \tag{6d}$$

$$v_{\mathrm{res}} = \frac{\lambda}{2T_f} \tag{6e}$$

$$v_{\max} = \frac{\lambda}{4T} \tag{6f}$$

where $\lambda$ is the wavelength of the transmitted signal.

The received signal model incorporates both array geometry and wave propagation physics. As shown in Fig. 3, for an $M \times N$ planar array with half-wavelength spacing ($d_x = d_y = \lambda/2$), the phase difference between adjacent elements depends on the angle of the incident wave. At each snapshot time $t$, the signal received at the $(m, n)$th element from the $i$th source is modeled as follows:

$$y_{m,n}(t) = \alpha_i s_i(t) \cdot e^{j2\pi\left(m\frac{d_x}{\lambda}\xi_i + n\frac{d_y}{\lambda}\zeta_i\right)} + \eta_{m,n}(t) \tag{7}$$

where $\xi_i = \sin\theta_i \cos\phi_i$ denotes the horizontal spatial frequency, $\zeta_i = \sin\theta_i \sin\phi_i$ denotes the vertical spatial frequency, and $\theta_i$ and $\phi_i$ represent the elevation and azimuth angles, respectively. $\alpha_i$ accounts for both path loss and phase shift, while $\eta_{m,n}(t)$ represents additive noise.

The array response vector for a hypothetical direction $(\theta, \phi)$ is given by the Kronecker product

$$\mathbf{a}(\theta, \phi) = \mathbf{a}_x(\xi) \otimes \mathbf{a}_y(\zeta) \tag{8}$$

with the subvectors along the $x$- and $y$-axes defined as

$$\left[\mathbf{a}_x(\xi)\right]_m = e^{j2\pi m(d_x/\lambda)\xi}, \quad \left[\mathbf{a}_y(\zeta)\right]_n = e^{j2\pi n\left(d_y/\lambda\right)\zeta}. \tag{9}$$

Here, $[\mathbf{a}_x(\xi)]_m$ and $[\mathbf{a}_y(\zeta)]_n$ implement the phase compensation required for coherent combining in directions $\xi$ and $\zeta$, respectively.

The angular power distribution is obtained via Capon's minimum variance distortionless response (MVDR) beamformer

$$P(\theta, \phi) = \frac{1}{\mathbf{a}^H(\theta, \phi)\mathbf{R}^{-1}\mathbf{a}(\theta, \phi)} \tag{10}$$

where $\mathbf{R} = \mathbb{E}[\mathbf{y}\mathbf{y}^H]$ is the signal covariance matrix. Inverse $\mathbf{R}^{-1}$ whitens the interference signals, creating deep nulls in their directions.

The maxima of the spatial spectrum reveal the elevation angles $\theta$ and azimuth angles $\phi$ through a grid search

$$\left(\hat{\xi}, \hat{\zeta}\right) = \arg\max_{\xi,\zeta} P(\theta, \phi). \tag{11}$$

The combined spatial frequencies encode the elevation angle $\theta$ through the Pythagorean identity

$$\theta = \arcsin\left(\sqrt{\hat{\xi}^2 + \hat{\zeta}^2}\right). \tag{12}$$

As illustrated in Fig. 3, the height $H$ of the detected object relative to the radar can be derived through transformation from spherical coordinates to Cartesian coordinates and is obtained according to the following expression:

$$H = R \cdot \cos\theta. \tag{13}$$

## III. Proposed System

### A. System Overview

The mmMotion system consists of six essential stages, as depicted in Fig. 4. These stages encompass data collection, feature extraction, a CNN–LSTM model, a real-time classifier, a classification result layer, and a smart appliances control module. The workflow of the system begins with acquiring human motion data using a 60-GHz mmWave radar. The radar sensor emits electromagnetic waves and captures the reflected signals from the human subject. These signals contain spatial information on the detected objects that can be utilized to characterize human motion. Subsequently, the raw radar data undergoes several signal processing steps to eliminate noise, clutter, and other undesired components. Afterward,

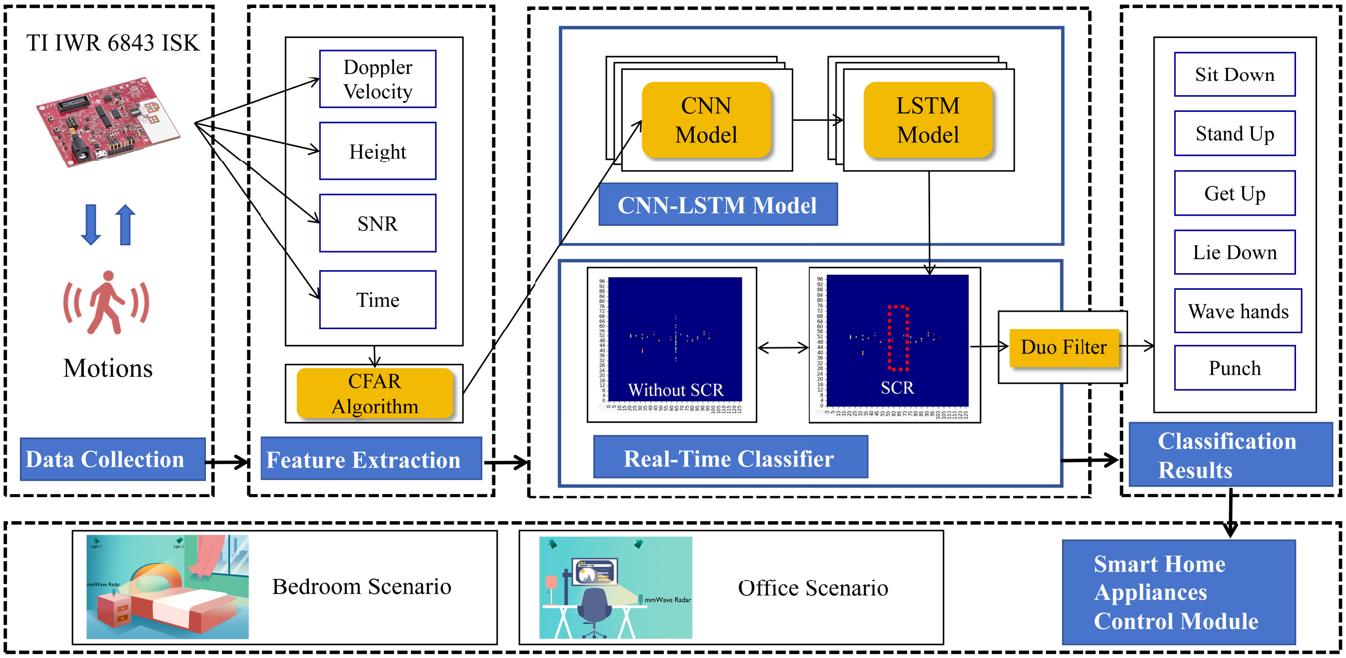



Fig. 4. Overview of the mmMotion system.

TABLE I
FMCW RADAR SETTING

| Parameters | Value |
|---|---|
| Working Frequency | 60-64 GHz |
| No. of RX Antennas | 4 |
| No. of TX Antennas | 3 |
| Slope of the Frequency $S$ | 66.58 MHz/us |
| Frame Duration | 50 ms |
| Range Resolution $R_{\mathrm{res}}$ | 0.044 m |
| Maximum Range $R_{\max}$ | 1.5 m |
| Velocity resolution $v_{\mathrm{res}}$ | 0.13 m/s |
| Maximum Velocity $v_{\max}$ | 6 m/s |

the processed data are fed into a feature extraction module that aims to extract pertinent features capable of effectively representing human motion. These extracted features are then employed for training a classifier responsible for recognizing and categorizing different types of human motion. To enable accurate recognition in real-time scenarios, a Real-time Classifier module is utilized. Ultimately, the output motion results obtained by the mmMotion system will facilitate controlling various smart appliances.

## B. Data Collection

We employ the Texas Instruments (TI) IWR 6843 ISK development board, which features an antenna array with four receive and three transmit antennas, to capture reflected signals from the target under observation. In configuring the mmWave radar for our indoor human motion detection system, the parameters are meticulously optimized to align with the specific detection requirements of bedroom and office environments, as detailed in Table I. The maximum detection range $R_{\max}$ is set to 1.5 m to minimize potential background interference. The range resolution $R_{\mathrm{res}}$ is configured to 0.044 m to enhance the capability of the DHT feature maps in capturing detailed characteristics of human motion. Additionally, the maximum detectable velocity $v_{\max}$ is set to be 6 m/s, with a velocity resolution $v_{\mathrm{res}}$ of 0.13 m/s, both chosen to fulfill the requirements of the intended detection tasks.

## C. Feature Extraction

The novel feature representation, DHT feature mapping, has been developed to effectively characterize and differentiate various human motions by capturing the temporal distribution of the detected object's Doppler velocity and height information. The Doppler velocity refers to the velocity of the detected points relative to the observer (radar), which is different from the real velocity in the physical world.

The Doppler velocity is normalized to a range of [0, 128] (#), where a value of 64 indicates stationary points relative to the radar. Values greater than 64 represent movement toward the radar, while values less than 64 indicate movement away from the radar. Similarly, the height $H$ is normalized to [0, 100] (#), with a value of 50 corresponding to points at the radar's level, values above 50 indicating the detected points above the radar, and values below 50 representing the detected points below the radar level.

The signal-to-noise ratio (SNR) is a measure used to quantify the level of a desired signal relative to the level of background noise. The SNR value is normalized to [0, 1] (#) mapping with different colors. When the SNR value approaches 1 (red), it indicates a stronger point relative to the noise signal and a closer proximity to the true location. Conversely, when the SNR value approaches 0 (blue), it signifies a weaker point relative to the noise signal and lower confidence in its proximity to the true location.

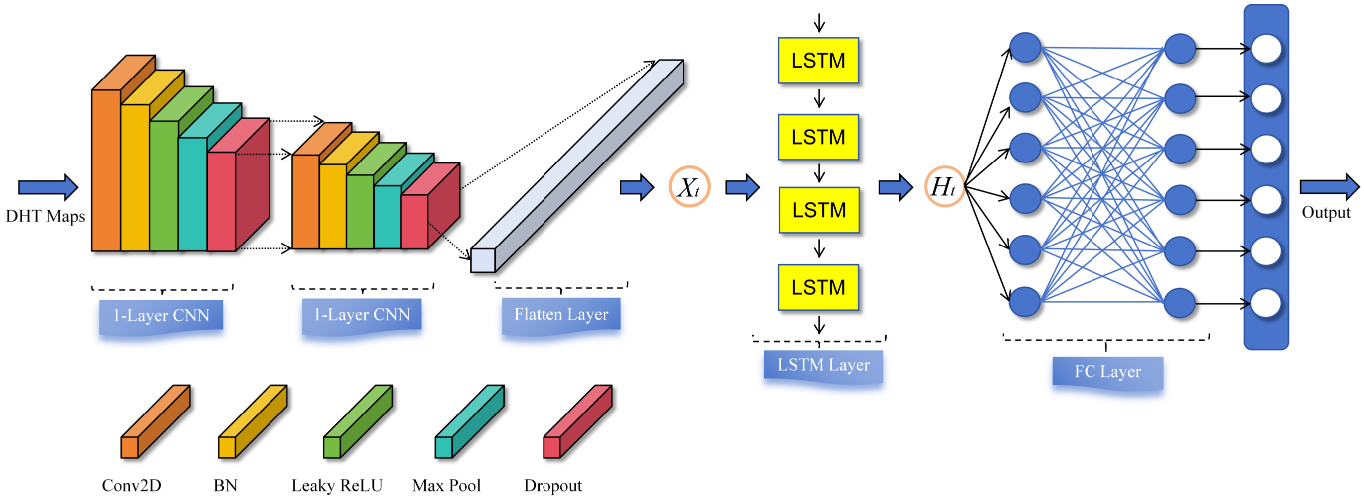



Fig. 5. Structure of the CNN–LSTM model.

The constant false alarm rate (CFAR) algorithm allows radar systems to adaptively set detection thresholds based on the local noise environment, ensuring reliable target detection even in varying conditions. In the mmMotion system, we utilize the CFAR algorithm to efficiently analyze and leverage radar cube data to greatly minimize data volume, allowing for streamlined transmission and supporting real-time applications. The CFAR threshold is experimentally determined to be 20 dB. The data compression performance of CFAR is evaluated in Section IV-G.

### D. CNN–LSTM Model

As illustrated in Fig. 5, the mmMotion framework integrates the CNN and LSTM architectures. The CNN component functions as the spatial modeling element of our framework. It is specifically designed to identify and extract spatial features and patterns from the data. The architecture includes convolutional layers, pooling layers, and nonlinear activation functions. The convolutional layers conduct operations with local receptive fields, enabling the model to detect spatial patterns effectively. Pooling layers help to reduce the spatial dimensions while preserving essential information. Nonlinear activation functions introduce complexities, allowing the model to learn intricate spatial representations. By integrating CNN into our framework, we can proficiently capture spatial relationships and derive discriminative features from the data, which enhances the model's ability to understand spatial contexts and make precise predictions based on visual patterns.

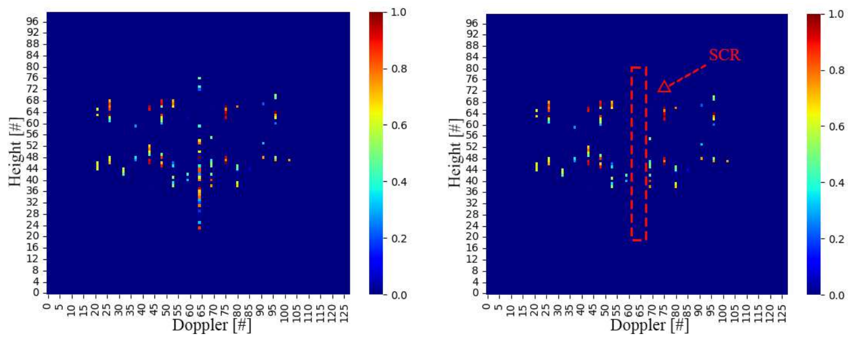
On the other hand, the LSTM component acts as the temporal modeling part of our framework. It is designed to capture and model sequential dependencies within the data. The LSTM architecture is composed of memory cells, input gates, forget gates, and output gates. Memory cells enable the model to retain information over extended periods, making it adept at capturing long-term dependencies. Input gates manage the flow of information into the memory cells, while forget gates decide which information to discard. Output gates regulate the values produced by the LSTM cells. By including LSTM in our model, we can effectively recognize temporal patterns and dependencies, allowing the model to learn from long-range contexts and make accurate predictions based on historical data.

The CNN and LSTM components are integrated by routing the output of the CNN into the LSTM. This combination allows the model to simultaneously capture both spatial and temporal features, taking advantage of the strengths of each architecture. All parameters are determined through experimental validation and training processes. Below is a comprehensive explanation of the model.

*1) CNN Cell:* We employ a two-layer CNN that extracts spatial features from sequential input data, such as a sequence of DHT feature maps.

1) CNN cell1 consists of the following.
    a) A depthwise separable convolution layer (groups = 4) with kernel size $3 \times 3$, stride 1, and padding 1, mapping 40 input channels to 60 output channels.
    b) A batch normalization (BN) layer over 60 channels.
    c) Leaky ReLU activation with negative slope of 0.01.
    d) $2 \times 2$ max-pooling with stride 2.
    e) A dropout layer with a probability of 0.05.
2) CNN cell2 maintains the same structure but reduces dimensionality.
    a) A convolution layer with identical kernel parameters, mapping 60 input channels to four output channels.
    b) A BN layer over four channels.
    c) Same activation, pooling, and dropout configuration as cell1.

*2) Flattening Layer:* This layer transforms the four-channel feature maps extracted by the CNN modules from a 3-D tensor of shape (batch, channels, and features) into a 1-D vector by flattening along the feature dimension (starting from dimension 2), producing an 800-D vector that serves as input to the LSTM cell.




(a) (b)

Fig. 6. DHT feature map (a) without SCR and (b) with SCR, using "punch" DHT feature maps as examples.

*3) LSTM Cell:* The LSTM cell is the core component of the system model. It is implemented as a standard LSTMCell module containing two learnable transformations.

1) *Input-to-Hidden (x2h):* A fully connected layer mapping the 800-D input features to a 1024-D hidden representation.
2) *Hidden-to-Hidden (h2h):* A recurrent connection mapping the 256-D previous hidden state to a 1024-D current state.

The cell maintains a hidden state size of 256 throughout the sequence processing.

*4) Output Layer:* A fully connected linear layer maps the 256-D hidden state from the LSTM cell to the final 6-D output vector, corresponding to six designated motions. The layer is implemented as

$$\text{output} = W_{256\times 6}h + b_6 \tag{14}$$

where $h$ is the final hidden state, $W$ is the learnable weight matrix, and $b$ is the bias term.

The integration of the CNN and LSTM offers several benefits. First, it enables a more comprehensive analysis of the data by considering both spatial and temporal aspects. This is particularly advantageous in tasks where understanding the spatial and temporal context is crucial. Second, the combined model can capture complex patterns and dependencies more effectively, leading to improved accuracy and robustness. Lastly, the end-to-end training of the combined model reduces the need for manual feature engineering and facilitates joint optimization of both spatial and temporal components.

Overall, our proposed model, which combines the CNN and LSTM architectures, offers enhanced data analysis capabilities by effectively capturing both spatial and temporal features. The integration of these architectures provides a synergistic effect, leading to improved performance and deeper insights into the underlying patterns within the data.

### *E. Real-Time Classifier*

*1) Static Clutter Removal:* We employ a coupled-model strategy during the training phase, utilizing two distinct classifiers trained on different input data representations. The first model uses standard DHT feature maps, while the second model is trained on feature maps that have undergone the static clutter removal (SCR) process, which filters out objects with zero Doppler velocity. The DHT feature maps and DHT–SCR feature maps are illustrated in Fig. 6(a) and (b). The original dataset is defined as $\boldsymbol{M} \in \mathbb{R}^{i\times j}$, where $i \in \{1, 2, \ldots, n\}$ is the serial number of chirp and spatial information is stored in column $j \in \{1, 2, \ldots, m\}$. Assuming the Doppler velocity $v$ is in the $k$th column, the reconstructed dataset $\boldsymbol{M}_*$ after SCR can be expressed as

$$\boldsymbol{M}_* = \left\{M[i]\left[j\right] \middle| M[i][k] \neq 0\right\} \tag{15}$$

where $\boldsymbol{M}[i][j]$ represents the element in the $i$th row and $j$th column.

This approach allows us to leverage the complementary strengths of both models, enhancing sensitivity to target motions and reducing false detections of random, undefined human movements. We combine the motion predictions from both models to make a final real-time output result. The coupled-model approach leverages the fact that the two models exhibit different behavior when presented with nontarget activities, such as oscillating motion with a similar magnitude to the target activities. After considering the predictions from both models, we can effectively reduce the chances of incorrectly recognizing nontarget motions as one of the desired poses, thereby improving the overall real-time detection accuracy.

*2) Duo Filter:* We utilize a sliding window dual-classifier methodology that continuously analyzes incoming radar frames, yielding motion classification results on an individual frame basis. This real-time detection framework operates by buffering a predetermined number of consecutive frames, specifically set to 40 frames, which corresponds to a duration of 2 s. The comprehensive logic flow is encapsulated in Algorithm 1. The prediction results of model training by DHT feature maps and by DHT–SCR feature maps are defined as $\text{predict}_0$ and $\text{predict}_1$, respectively. The six designated motions (sit down, punch, wave hands, get up, stand up, and lie down) are correspondingly categorized by $\Psi \in \{\alpha, \beta, \gamma, \delta, \epsilon, \zeta\}$. When the recognition times of the two models $\text{predict}_0$ and $\text{predict}_1$ simultaneously exceed the threshold of the specified Duo Filter, the real-time detection system verifies the existence of the motion and outputs the result $\Psi$. The term $\text{predict}_n[-i]$ denotes the $i$th most recent prediction generated by the corresponding model. As illustrated in Algorithm 1, the system will present $\Psi = \alpha$ when all of the preceding eight consecutive predictions

**Algorithm 1** Duo Filter

**Require:** DHT model predict result $predict_0$ and DHT-SCR model predict result $predict_1$

**Ensure:** Real-time motion detection result $\Psi \in \{\alpha, \beta, \gamma, \delta, \epsilon, \zeta, \eta\}$

**procedure** $(predict_0, predict_1, \Psi)$
  **if** $\forall i \in \{1, 2, \ldots, 8\}$, $predict_0[-i] == \alpha \land predict_1[-i] == \alpha \land predict_1[-9] == \alpha$ **then**
    $\Psi = \alpha \leftarrow Sitdown$
  **else if** $\forall i \in \{1, 2\}$, $predict_0[-i] == \beta \land predict_1[-i] == \beta \land predict_1[-3] == \beta$ **then**
    $\Psi = \beta \leftarrow Punch$
  **else if** $\forall i \in \{1, 2\}$, $predict_0[-i] == \gamma \land predict_1[-i] == \gamma \land predict_1[-3] == \gamma$ **then**
    $\Psi = \gamma \leftarrow Wavehands$
  **else if** $\forall i \in \{1, 2, 3, 4\}$, $predict_0[-i] == \delta \land predict_1[-i] == \delta$ **then**
    $\Psi = \delta \leftarrow Getup$
  **else if** $\forall i \in \{1, 2, 3, 4, 5, 6\}$, $predict_0[-i] == \epsilon \land predict_1[-i] == \epsilon \land predict_1[-7] == \epsilon$ **then**
    $\Psi = \epsilon \leftarrow Standup$
  **else if** $\forall i \in \{1, 2, 3, 4, 5\}$, $predict_0[-i] == \zeta \land predict_1[-i] == \zeta$ **then**
    $\Psi = \zeta \leftarrow Liedown$
  **else**
    $\Psi = \eta \leftarrow Still$
  **end if**
**end procedure**

(predict$_0$) and nine consecutive predictions (predict$_1$) are equal to $\alpha$. If none of the six designated motions are detected, the system will display the static pose $\Psi = \eta$. The Duo Filter threshold values are individually tailored for each of the six designated motions, based on empirical observations and experimental evaluations. This strategy helps to minimize the effects of sporadic misclassifications, ensuring more reliable motion detection.

### *F. Smart Home Appliances Control Module*

Finally, we design a smart appliance control module for the mmMotion system that translates motion detections into specific home automation commands. When the real-time classifier identifies one of the six designated motions $\Psi \in \{\alpha, \beta, \gamma, \delta, \epsilon, \zeta\}$, it triggers corresponding smart home motions through the MQTT protocol as detailed in Algorithm 2. The system maps sitting down $\alpha$ to turning on the table lamp, standing up $\epsilon$ to turning it off, while a punching motion $\beta$ activates the ceiling light. For bedroom scenarios, getting up from bed $\delta$ switches the lighting to night mode (gradual brightening) and lying down $\zeta$ activates day mode (gradual dimming). Additionally, waving hands $\gamma$ controls the curtains to open. The mmWave radar connects via USB to a laptop (HP Victus 15L with an Intel i7-12700H processor and 16GB RAM) that handles both the classification algorithm and MQTT communication with the smart home hub through Wi-Fi. While the implementation details of the control protocol (including MQTT topic structure and error handling mechanisms) extend

**Algorithm 2** Smart Appliances Control System

**Require:** Real-time Classifier output $\Psi \in \{\alpha, \beta, \gamma, \delta, \epsilon, \zeta\}$

**Ensure:** Appliance states and scene changes

**if** $\Psi = \epsilon$ **then** ▹Stand up
  `mqtt_publish(''lamp/command'',''off'')`
  `log_action(''Lamp off'')`
**else if** $\Psi = \alpha$ **then** ▹Sit down
  `mqtt_publish(''lamp/command'',''on'')`
  `log_action(''Lamp on'')`
**else if** $\Psi = \beta$ **then** ▹Punch
  `mqtt_publish(''light/command'',''on'')`
  `log_action(''Light on'')`
**else if** $\Psi = \gamma$ **then** ▹Wave hands
  `mqtt_publish(''curtain/command'',''open'')`
  `log_action(''Curtain open'')`
**else if** $\Psi = \delta$ **then** ▹Get up
  `mqtt_publish(''lighting/mode'',''night'')`
  `log_action(''Night mode'')`
**else if** $\Psi = \zeta$ **then** ▹Lie down
  `mqtt_publish(''lighting/mode'',''day'')`
  `log_action(''Day mode'')`
**end if**

beyond this article's scope, our demo video demonstrates the complete system workflow: https://youtu.be/0z10HDJb2XU.

## IV. Experimental Results

In this section, we present our experimental configuration, assessment criteria, and outcomes achieved using our proposed mmWave radar-based system for recognizing human motions. A comprehensive set of experiments is carried out to evaluate the offline and real-time testing performance of our model in accurately detecting and categorizing various human motions.

### *A. Experiment Implementation*

To assess the effectiveness of the mmMotion system in accurately identifying human motion, we perform a comprehensive data collection process within a controlled laboratory setting. The study involves 22 participants who are instructed to execute six predefined motions: sitting down (on a chair), standing up (from a chair), getting up (from a bed), lying down (on a bed), waving their hands (in the air), and punching (in the air). Furthermore, participants are directed to maintain a stationary pose or engage in minimal movements. Each participant repeats each motion multiple times, with each cycle lasting between 1 to 2 s. All participants are university students between the ages of 22 and 29, representing a diverse sample of gender, height, weight, body mass index (BMI), and clothing. Key factors of the experimental subjects are recorded in Table II.

The mmWave radar sensor is positioned in accordance with the diagram depicted in Fig. 7. During the data collection process, the radar is situated at a distance of $R_d = 70 \sim 100$ cm from the participant's main body, elevated to a height of $H_d = 100 \sim 120$ cm above ground level. To ensure consistency and facilitate accurate motion recording, participants are seated

TABLE II
DATA ACQUISITION OF VARIOUS EXPERIMENTAL SUBJECTS

| **Key Factor** | **Subject Number** | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | **1** | **2** | **3** | **4** | **5** | **6** | **7** | **8** | **9** | **10** | **11** | **12** | **13** | **14** | **15** | **16** | **17** | **18** | **19** | **20** | **21** | **22** |
| **Gender** | $M$ | $M$ | $M$ | $M$ | $F$ | $M$ | $M$ | $M$ | $M$ | $M$ | $M$ | $M$ | $M$ | $M$ | $F$ | $M$ | $M$ | $M$ | $F$ | $F$ | $M$ | $M$ |
| **Age** | 23 | 26 | 23 | 27 | 25 | 23 | 26 | 22 | 30 | 22 | 23 | 27 | 24 | 23 | 26 | 29 | 24 | 27 | 23 | 23 | 24 | 23 |
| **Height (cm)** | 175 | 176 | 176 | 185 | 163 | 175 | 184 | 175 | 185 | 181 | 166 | 176 | 155 | 185 | 168 | 172 | 171 | 173 | 165 | 165 | 178 | 170 |
| **Weight (Kg)** | 72 | 72 | 64 | 76 | 44 | 69 | 92 | 63 | 105 | 82 | 65 | 66 | 48 | 71 | 54 | 85 | 50 | 75 | 53 | 58 | 68 | 90 |
| **BMI** | 24 | 23 | 21 | 22 | 17 | 23 | 27 | 21 | 31 | 25 | 24 | 21 | 20 | 20 | 19 | 29 | 17 | 25 | 20 | 21 | 21 | 31 |
| **Clothing*** | $A$ | $A$ | $A$ | $B$ | $C$ | $A$ | $A$ | $C$ | $A$ | $A$ | $C$ | $A$ | $A$ | $A$ | $A$ | $A$ | $A$ | $A$ | $B$ | $C$ | $A$ | $A$ |

*Type of clothing: $A$ refer to wearing cotton T-shirt; $B$ refer to wearing sweater; $C$ refer to wear jacket.

TABLE III
OFFLINE MODEL TRAINING RESULTS AND COMPUTATIONAL METRICS

| **Model** | **Accuracy (%)** | **Parameters (M)** | **FLOPs (M)** | **Time Delay (ms)** |
|---|---|---|---|---|
| CNN only | 76.47 | 0.29 | 3.87 | 32 |
| LSTM only | 85.12 | 0.56 | 1.98 | 38 |
| 1-layer CNN + 1-layer LSTM | 96.05 | 0.84 | 6.31 | 42 |
| 1-layer CNN + 2-layer LSTM | 96.67 | 0.97 | 7.02 | 45 |
| **2-layer CNN + 1-layer LSTM** | **98.13** | **1.09** | **7.34** | **48** |
| 2-layer CNN + 2-layer LSTM | 96.69 | 1.53 | 12.77 | 103 |
| 3-layer CNN + 1-layer LSTM | 84.19 | 1.96 | 19.43 | 128 |
| 3-layer CNN + 2-layer LSTM | 78.45 | 2.34 | 23.89 | 154 |

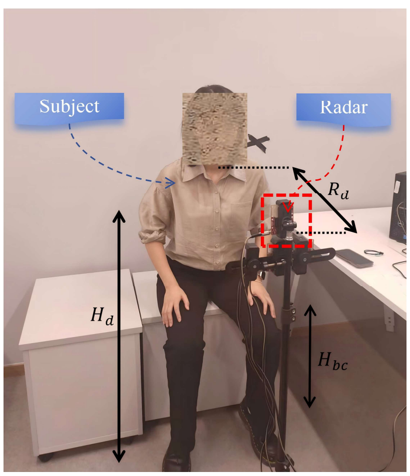



Fig. 7. Experimental setting for mmWave radar data collection.

on a chair or a bed measuring $H_{bc}$ = 40 ~ 60 cm in height. The distance and height parameters are set in accordance with the actual settings in the office and bedroom scenarios. The radar frame rate is configured to capture motion dynamics at a frequency of 20 frames/s (20 Hz), thereby providing an optimal temporal resolution for subsequent analysis.

## *B. Offline Model Training Results*

We conduct offline training and evaluation using a dataset comprising 52 280 radar data samples capturing human motions, encompassing motions such as sitting down on a chair (8720), standing up from a chair (8840), getting up from a bed (7360), lying down on a bed (7580), waving hands (9120), and punching (9660). The dataset is randomly partitioned into two subsets for training and evaluation, with 75% allocated to training and the remaining 25% for evaluation. Throughout the training phase, we implement the Adam optimizer using a learning rate of 0.001 and adopt a batch size of 60. The training process spans over 50 epochs, ultimately selecting the model that demonstrates the highest validation accuracy for further evaluation.

Our comprehensive evaluation of different model architectures reveals several key insights (see Table III). The standalone CNN and LSTM models achieve modest accuracy (76.47% and 85.12%, respectively), demonstrating their individual limitations in capturing both spatial and temporal features. Hybrid architectures show significant improvement, with the one-layer CNN + one-layer LSTM combination reaching 96.05% accuracy. Notably, the two-layer CNN + one-layer LSTM configuration achieves the optimal balance between performance and efficiency, delivering 98.13% accuracy with 1.09 M parameters and 7.34 M floating-point operations (FLOPs), while maintaining real-time capability of 48-ms latency. Deeper architectures (three-layer variants) show diminishing returns, with accuracy dropping below 85% despite increased computational cost (up to 23.89 M FLOPs and 154 ms latency), suggesting overfitting and excessive model complexity. These results confirm our architectural choice for the mmMotion system, where the two-layer CNN + one-layer LSTM structure provides the best tradeoff between recognition accuracy and computational efficiency for real-time deployment.

## *C. Different Feature Extraction Methods Evaluation*

To better illustrate the mapping relationship between different feature extraction methods and various motions, we have generated DHT feature maps, RDT feature maps, and elevation ADT feature maps for each of the six designed motions. To

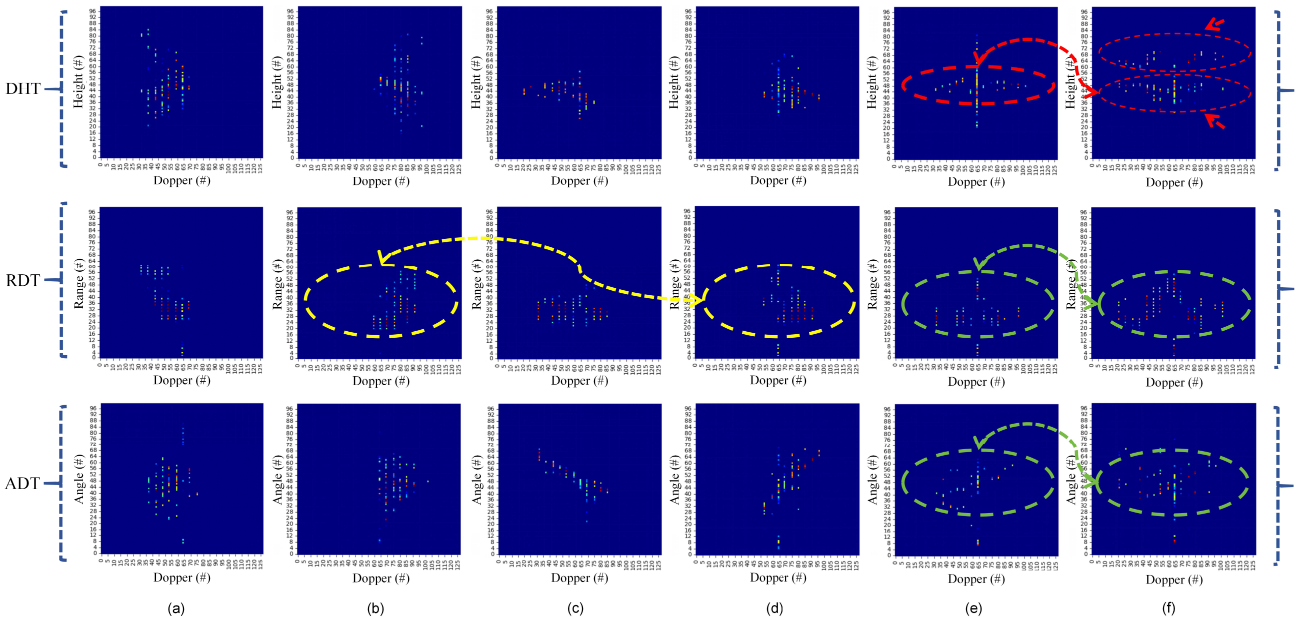



Fig. 8. DHT, RDT, and ADT feature maps comparison of motion. (a) Sit down. (b) Stand up. (c) Get up. (d) Lie down. (e) Wave hands. (f) Punch.

facilitate horizontal comparison, we have scaled the transverse Doppler velocity values to a range of [0, 128] (#), while converting the vertical coordinates of height $H$, range $R$, and elevation angle $\theta$ in all three figures to a range of [0, 100] (#). The colors in the point cloud represent the magnitude of the SNR value.

Compared with ADT, DHT incorporates the range factor. In addition, compared with RDT, DHT incorporates the elevation angle factor. By fully exploiting information from 5-D—range $R$, elevation angle $\theta$, Doppler velocity $v$, SNR, and time—DHT enables a more accurate representation of the spatial characteristics of motions. The DHT feature maps of "wave hands" and "punch" motions reveal distinct kinematic signatures that enable effective discrimination between these motions. As shown in the red circle in Fig. 8, the punch motion exhibits two characteristic curves in the DHT map, corresponding to the differential height variations of the hand and elbow during the punching motion. In contrast, the wave hands motion produces only a single oscillating curve, reflecting more uniform height changes, as shown in red arrows in Fig. 8. This discriminative capability stems from DHT's explicit encoding of height-velocity coupling, which captures subtle spatial-temporal patterns that are lost in conventional representations. Notably, RDT and ADT maps fail to distinguish between these motions, as both exhibit similar single-curve patterns in these representations. This limitation occurs because both waving and punching produce comparable range and elevation angle variations, making them indistinguishable when height information is absent, as shown in the green circle in Fig. 8.

To ensure statistical robustness, we performed fivefold cross-validation on the dataset. The results, summarized in Table IV, show that DHT features consistently outperform RDT and ADT across all folds, with an average accuracy of 98.13% $\pm$ 0.18%, compared to 86.22% $\pm$ 0.20% for RDT and 91.51% $\pm$ 0.19% for ADT. Moreover, models using RDT and ADT inputs achieve significantly lower accuracy for these motions (67.86% for RDT-wave hands, 74.59% for RDT-punch, 79.37% for ADT-wave hands, and 83.89% for ADT-punch). The superiority of DHT features extends beyond upper-body motions. For whole-body movements like standing up and lying down, RDT maps show poor discrimination (89.71% and 90.62% accuracy, respectively), as these motions generate similar range variations relative to the radar, as shown in the yellow circle in Fig. 8. In contrast, DHT maps clearly differentiate these motions through their distinct height transition patterns. Our comprehensive evaluation demonstrates that DHT feature maps consistently outperform alternative representations across all motion categories. This advantage is particularly pronounced for motions involving significant height variations, where DHT's spatial encoding provides essential discriminative information that is absent in RDT and ADT representations. The robust performance across both large-scale (sit down, stand up, get up, lie down) and fine-grained (wave hands, punch) motions confirms DHT's effectiveness as the optimal input representation for our mmMotion system.

### D. Real-Time Classifier Evaluation

We conduct a real-time test experiment to evaluate the real-time detection rate of mmMotion by the real-time classifier in office scenarios and bedroom scenarios. The experiment involves five participants who are instructed to perform assigned tasks in the presence of mmWave radar. As depicted in Fig. 9, the participants are required to execute six predetermined motions (Sitting down, standing up, getting up, lying down, waving hands, and punching) under controlled conditions. Following each motion, there is a 2-second interval before proceeding to the subsequent motion. A real-time detection system is deemed accurate if it yields the correct outcome within one second after the completion of each movement by the participant. Each individual performs each motion a total of 20×.

TABLE IV
*k*-FOLD CROSS-VALIDATION RESULTS FOR DIFFERENT FEATURE EXTRACTION METHODS ACROSS ALL MOTIONS

| Motion | Feature Type | Fold 1 | Fold 2 | Fold 3 | Fold 4 | Fold 5 | Mean ± Std |
|---|---|---|---|---|---|---|---|
| Sit down | DHT | 97.10% | 97.50% | 97.60% | 97.30% | 97.65% | **97.43% ± 0.23%** ⋆ |
| | RDT | 96.50% | 96.90% | 96.80% | 96.60% | 96.90% | 96.74% ± 0.18% |
| | ADT | 97.90% | 98.30% | 98.20% | 97.95% | 98.30% | 98.13% ± 0.19% † |
| Stand up | DHT | 98.30% | 98.80% | 98.70% | 98.50% | 98.80% | **98.63% ± 0.22%** ⋆ |
| | RDT | 89.40% | 89.90% | 89.75% | 89.55% | 89.85% | 89.71% ± 0.21% † |
| | ADT | 91.95% | 92.45% | 92.30% | 92.10% | 92.35% | 92.23% ± 0.20% |
| Get up | DHT | 96.80% | 97.20% | 97.15% | 96.90% | 97.05% | **97.02% ± 0.16%** ⋆ |
| | RDT | 97.65% | 98.00% | 97.85% | 97.70% | 97.95% | 97.83% ± 0.15% |
| | ADT | 98.70% | 99.05% | 98.90% | 98.75% | 98.95% | 98.87% ± 0.15% † |
| Lie down | DHT | 97.75% | 98.15% | 98.10% | 97.85% | 98.05% | **97.98% ± 0.16%** ⋆ |
| | RDT | 90.40% | 90.80% | 90.65% | 90.45% | 90.80% | 90.62% ± 0.18% † |
| | ADT | 96.20% | 96.60% | 96.45% | 96.25% | 96.55% | 96.41% ± 0.18% |
| Wave hands | DHT | 98.35% | 98.75% | 98.65% | 98.45% | 98.65% | **98.57% ± 0.16%** ⋆ |
| | RDT | 67.65% | 68.05% | 67.90% | 67.70% | 68.10% | 67.86% ± 0.20% † |
| | ADT | 79.15% | 79.55% | 79.40% | 79.20% | 79.55% | 79.37% ± 0.19% |
| Punch | DHT | 98.95% | 99.35% | 99.25% | 99.05% | 99.20% | **99.16% ± 0.16%** ⋆ |
| | RDT | 74.35% | 74.75% | 74.60% | 74.40% | 74.85% | 74.59% ± 0.21% † |
| | ADT | 83.65% | 84.05% | 83.90% | 83.70% | 84.15% | 83.89% ± 0.22% |
| **Overall** | DHT | 97.90% | 98.30% | 98.25% | 98.00% | 98.20% | **98.13% ± 0.18%** ⋆ |
| | RDT | 85.95% | 86.35% | 86.20% | 86.00% | 86.40% | 86.22% ± 0.20% † |
| | ADT | 91.25% | 91.65% | 91.50% | 91.30% | 91.65% | 91.51% ± 0.19% |

⋆ Best performance; † Lowest performance among methods.

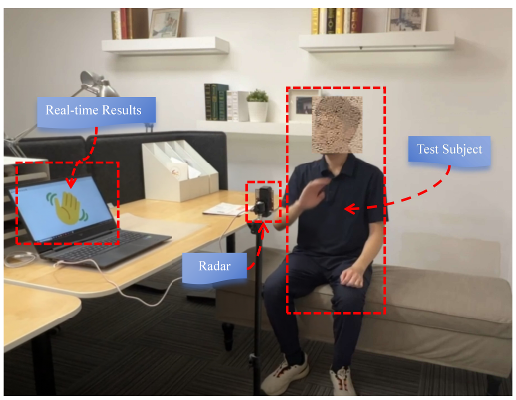



Fig. 9. Real-time testing environment for evaluating the performance of mmMotion in a smart home scenario.

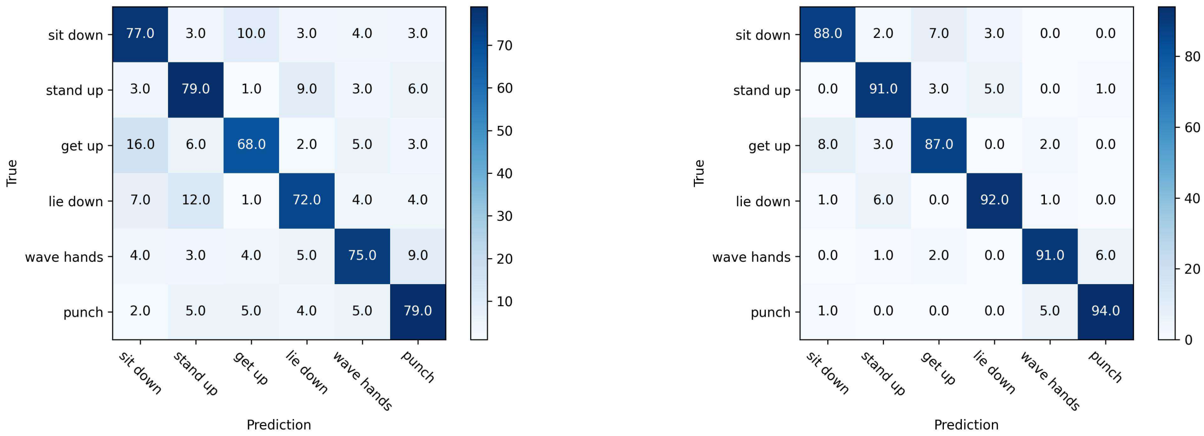
The confusion matrices are presented in Fig. 10(a) and (b), illustrating the real-time classification results when comparing the usage of a Real-time Classifier with its absence. The *x*- and *y*-axes of the confusion matrix correspond to the predicted result and true labels of human motion. The confusion matrix reveals a significant enhancement in the real-time accuracy of each motion following the adoption of the Real-time Classifier. The overall real-time recognition accuracy has been elevated from 75% to 90.5%, underscoring the indispensability of employing a real-time classifier for mmMotion.

It is worth noting that the recognition accuracy for sitting down and getting up motions is comparatively lower when compared to other motions. This discrepancy can be attributed to the predominantly negative Doppler velocities observed during sit-down and get-up movements (in comparison to distant radar movements), as well as minimal variations in relative height due to differences in subjects' heights and movements. Consequently, these factors collectively contribute to system misjudgment. On the other hand, the punch motion exhibits superior recognition performance and is less prone to being misidentified as other motions. The DHT feature maps reveal two distinct height curves during the punch motion, which result from the arm constantly being pulled back close to the torso. However, it should be noted that there may still exist confusion between the punch motion and the wave hands motion since both exhibit curved patterns on the DHT feature maps. When experimental subjects wave their hands differently, some individuals may also incorporate large arm and elbow movements, leading their wave motions to reflect two lines on the DHT feature maps. In summary, individual height differences and varying movement habits are identified as key factors influencing real-time detection rates of this system.

### *E. Real-Time Generalization and Robustness Test Evaluation*

On the one hand, due to individual differences in physical characteristics (height and weight) as well as the range and patterns of motion, real-time detection systems often fail to accurately recognize motions performed by new users who were not involved in the collection of the training dataset. Consequently, real-time systems exhibit lower generalization capability compared to conventional offline systems. On the other hand, the offline classifier is limited to displaying classification probabilities for the six pretrained motions. In contrast, real-time applications frequently encounter interference from untrained motions, as large-scale movements may contain kinematic features similar to those of smaller, trained motions. As a result, real-time detection systems are more prone to misclassifying interfering motions as one of the six predefined motions, thereby demonstrating reduced robustness when compared to the offline classification system.

To evaluate the real-time generalization and robustness of the mmMotion system across diverse individuals and environments, we enlist ten independent participants who are not involved in the data collection process to execute random movements in two distinct settings: a controlled laboratory




(a) (b)

Fig. 10. Confusion matrices for real-time motion recognition. (a) Without a real-time classifier. (b) With a real-time classifier.

TABLE V
GENERALIZATION AND ROBUSTNESS EVALUATION ACROSS CONTROLLED LABORATORY AND REAL HOME ENVIRONMENTS

| **Environments** | **Method** | **Large-Motion Interference Test** $\Phi_{\text{design}} + \Phi_{\text{large}}$ | | | | **Small-Motion Interference Test** $\Phi_{\text{designated}} + \Phi_{\text{small}}$ | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | *Acc* | *Precision* | *Recall* | $F1$-score | *Acc* | *Precision* | *Recall* | $F1$-score |
| **laboratory** | **Without Real-time Classifier** | 72.50% | 87.01% | 74.44% | 80.24% | 70.83% | 86.67% | 72.22% | 78.79% |
| | **With Real-time Classifier** | 89.17% | 95.29% | 90.00% | 92.57% | 86.67% | 93.02% | 88.89% | 90.91% |
| **Real Scenarios** | **Without Real-time Classifier** | 71.67% | 86.32% | 73.89% | 79.58% | 69.17% | 85.42% | 71.11% | 77.55% |
| | **With Real-time Classifier** | 88.33% | 94.44% | 89.17% | 91.72% | 85.00% | 92.11% | 87.22% | 89.60% |

environment and real-world smart home scenarios. Among these participants, 5 are tested in the controlled laboratory environment, while the other five are evaluated in real smart home scenarios. The designated motion set, denoted as $\Phi_{\text{design}}$, consists of six well-trained motions. The large-motion interference set, denoted as $\Phi_{\text{large}}$, comprises three nontrained motions involving significant movements, such as walking, bending over, and kicking. The small-motion interference set, denoted as $\Phi_{\text{small}}$, includes three nontrained motions with smaller movements, such as snapping, shaking the head, and nodding. In the experiment, each participant is randomly assigned to perform three trials of $\Phi_{\text{designated}}$ and one trial of $\Phi_{\text{large}}$. Subsequently, they are asked to perform three trials of $\Phi_{\text{designated}}$ and one trial of $\Phi_{\text{small}}$ in a random order. Misclassifying any of $\Phi_{\text{large}}$ or $\Phi_{\text{small}}$ motions as one of $\Phi_{\text{design}}$ would result in a false detection. Therefore, this test can be viewed as a binary classification problem, and we utilize accuracy rate (Acc), precision rate (Precision), recall rate (Recall), and $F1$-score ($F1$-score) metrics to evaluate the generalization ability and robustness of mmMotion system.

Based on the comprehensive evaluation results presented in Table V, the real-time classifier demonstrates significant performance enhancements across both testing environments. In the controlled laboratory setting, the incorporation of the real-time classifier yields a remarkable improvement of 16.67% in accuracy for large-motion interference tests and 15.83% for small-motion interference tests. Similarly, under real-world scenarios, the system maintains robust performance with accuracy improvements of 16.66% and 15.83% for large and small motion interference tests, respectively. The system exhibits superior robustness against large-motion interference compared to small-motion interference, as evidenced by the higher accuracy rates in both environments (89.17% vs. 86.67% in the laboratory; 88.33% vs. 85.00% in real scenarios). This performance differential can be attributed to the more distinct height variations characteristic of large movements, which provide clearer discriminative features for classification. Conversely, the greater variability in individual execution styles for small movements—particularly subtle differences in body trunk positioning or arm elevation—introduces additional challenges for motion discrimination. Notably, the minimal performance degradation observed between laboratory and real-world environments (approximately 1% difference) underscores mmMotion's exceptional generalization capability. The consistent maintenance of high precision and recall metrics across diverse testing conditions further validates the system's robustness against both environmental variations and individual movement differences.

### *F. Comparison to the State-of-the-Art Results*

To further demonstrate the superiority of the mmMotion system, we compare our method with state-of-the-art approaches. As presented in Table VI, mmMotion outperforms other methods by achieving the highest real-time classification performance. Notably, a significant number of existing state-of-the-art methods lack the capability for real-time motion detection.

TABLE VI
COMPARISON OF THE PROPOSED METHOD WITH STATE-OF-THE-ART RESULTS

| Method | Feature Input | Model | No. of Motions | Classification Accuracy | | Hardware |
|---|---|---|---|---|---|---|
| | | | | Off-line Training | Real-time Performance | |
| CRSFD [41] | Doppler-Angle-Time | CNN | 6 | 97.15% | \ | 5.8 GHz |
| PDHE-Net [42] | Doppler-Time | CNN | 6 | 96.67% | \ | 60 GHz |
| MAEF-Net [43] | Doppler-Time | CNN | 7 | 98.33% | \ | 60 Ghz |
| [44] | Frequency-Time | DCNN | 5 | 90.7% | \ | 7.25 GHz |
| [45] | Frequency-Time | 1-D-DAN | 6 | 98.11% | \ | 24 GHz |
| RadarNet [46] | End-to-End | CNN | 7 | 96.31% | \ | 24 GHz |
| CubeLearn [26] | End-to-End | CNN-LSTM | 6 | 98.06% | \ | 60 GHz |
| m-Activity [47] | Point Clouds | CNN-GRU | 5 | 93.25% | 89.07% | 77 GHz |
| RadHAR [29] | Point Clouds | CNN-LSTM | 5 | 90.47% | \ | 77 GHz |
| HRNet [48] | Point Clouds | cGAN | 6 | 95% | \ | 60 Ghz |
| **mmMotion** | **Doppler-Height-Time** | **CNN-LSTM** | **6** | **98.13%** | **90.5%** | **60 GHz** |

*1) Feature Extraction Methodology:* Existing works predominantly rely on Doppler–time (DT) spectrograms, such as CRSFD [41], PDHE-Net [42], and MAEF-Net [43], which achieve reasonable accuracy but neglect spatial height information critical for fine-grained motion discrimination. Frequency–time (FT) representations, as used by Kim and Moon [44] (90.7%) and Lai et al. [45] (98.11%), also lack 3-D spatial encoding. Point-cloud-based approaches like the HRNet [48] (95%) and m-Activity [47] (93.25%) process voxelized data but suffer from high computational overhead. In contrast, mmMotion introduces DHT feature maps, explicitly integrating RD and elevation-angle information to capture 3-D spatial–temporal dynamics. This innovation enables superior motion discrimination, contributing to our 98.13% offline accuracy—surpassing CRSFD (97.15%), PDHE-Net (96.67%), and closely rivaling the MAEF-Net (98.33%) while maintaining real-time capability.

*2) Network Architecture:* Prior works employ various architectures, including CNNs (e.g., CRSFD [41] and PDHE-Net [42]) and hybrid CNN–LSTMs (e.g., CubeLearn [26] and RadHAR [29]); however, many suffer from high computational complexity or lack real-time optimization. For instance, CubeLearn uses an end-to-end CNN–LSTM but fails to achieve real-time detection, and RadarNet [46] relies on learnable preprocessing, increasing latency. mmMotion adopts a lightweight CNN–LSTM optimized for edge deployment, where the CNN extracts hierarchical spatiotemporal features from DHT maps, and the LSTM captures temporal dependencies efficiently. Compared to m-Activity [47] (CNN–GRU, 89.07% real-time accuracy) and RadHAR (CNN–LSTM, 90.47% offline accuracy), our model achieves 90.5% real-time recognition with 48ms latency, making it ideal for IoT applications.

*3) Real-Time Performance:* A critical limitation of existing works is their inability to support real-time classification. For example, CRSFD [41] and PDHE-Net [42] report no real-time results, while m-Activity [47] achieves 89.07% real-time accuracy but requires DBSCAN clustering, increasing latency. The HRNet [48] focuses on static pose estimation (3.2 s/frame), rendering it unsuitable for dynamic motions. mmMotion addresses this gap with a real-time classifier incorporating SCR and a Duo Filter for adaptive thresholding and temporal smoothing, enabling 90.5% real-time accuracy—1.43% higher than m-Activity.

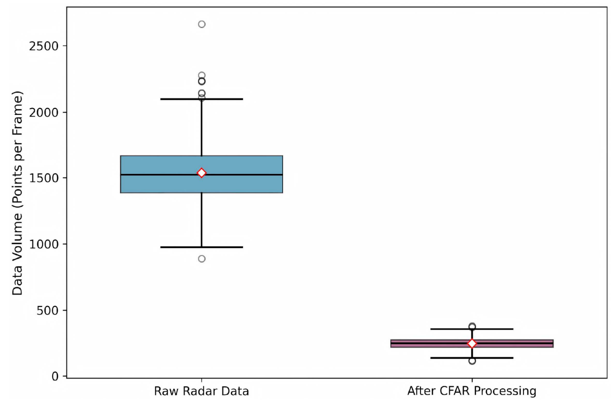



Fig. 11. CFAR data compression performance.

*4) Comparison With Attention-Based Frameworks:* Recent attention mechanisms, such as the MAEF-Net [43], improve feature discrimination but increase computational overhead. While the MAEF-Net achieves 98.33% accuracy using multi-attention fusion, it lacks real-time capability and height-aware features. mmMotion demonstrates that lightweight CNNs with LSTMs can rival attention-based models without sacrificing speed. Our DHT feature maps inherently encode motion saliency, reducing the need for auxiliary attention modules.

## G. Computational Efficiency and Power Consumption

To address the practical considerations of deployment, we analyze the computational complexity and power consumption of the mmMotion system running on the aforementioned laptop platform (HP Victus 15L with an Intel i7-12700H processor and 16GB RAM). While this platform is not representative of a final low-cost IoT device, it provides an upper-bound measurement of the algorithm's efficiency and a baseline for future optimization on edge hardware.

As illustrated in Fig. 11, the CFAR processing achieves an average data compression rate of 85.3%, reducing the raw radar data from approximately 1449 $\pm$ 225 points per frame to only 215 $\pm$ 43 points. This substantial data reduction (6.7:1 compression ratio) dramatically decreases the computational burden for subsequent feature extraction and model inference stages. The computational complexity of the core classification model is precisely quantified in Section IV-B at 7.34 M FLOPs per frame, with an inference latency of 48 ms, confirming

TABLE VII
POWER CONSUMPTION ANALYSIS ON THE LAPTOP DEPLOYMENT PLATFORM

| System State | Average Power (W) | Description |
|---|---|---|
| Laptop Idle (Baseline) | 45.2 | OS running, display on, radar connected but idle |
| Active Sensing & Classification | 48.5 | Real-time operation of mmMotion system |
| **Power Delta (mmMotion System)** | **+3.3** | Additional power consumed by mmMotion processing |

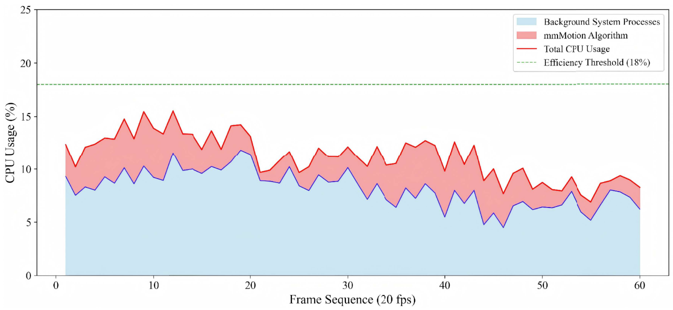



Fig. 12. Computational efficiency analysis.

the lightweight nature of the CNN–LSTM architecture. The MVDR beamformer, used for angle of arrival estimation, has a known complexity that scales with the number of antennas. However, in our application, this cost is mitigated by the low frame rate (20 Hz) and a constrained angular search space (120°), making it feasible for the real-time pipeline on a specialized radar signal-processing digital signal processor (DSP). As shown in Fig. 12, the algorithm maintains an average CPU usage of 4.2% during real-time operation, with total system consumption consistently below the 18% efficiency threshold.

The power consumption during operation is measured using a Joulescope JS220 hardware power monitor. The results, presented in Table VII, illustrate the system's power profile. The high baseline consumption is characteristic of a laptop. The marginal increase in power during active classification (a delta of +3.3 W) confirms the computational efficiency of the mmMotion algorithm itself. The primary contribution of this work is the validation of the high-accuracy, real-time algorithmic pipeline. The results demonstrate that the mmMotion system is sufficiently efficient to be a candidate for porting to a low-power edge device, which is essential for achieving the goal of a cost-effective, always-on smart home sensor.

## V. CONCLUSION

In conclusion, the proposed mmMotion system addresses critical challenges in real-time human motion detection within smart home IoT scenarios using mmWave radar. By leveraging the nonintrusive nature and privacy advantages of mmWave radar, mmMotion effectively recognizes six key human motions: sitting down, standing up, getting up, lying down, waving hands, and punching. Our novel feature extraction method utilizes the DHT feature maps, marking a significant advancement in the field. The integration of a lightweight CNN and LSTM architecture enables the system to simultaneously capture spatial and temporal features, thereby enhancing detection accuracy. Additionally, by implementing a real-time classifier along with the SCR method and Duo Filter algorithm significantly improves mmMotion's real-time classification performance. Experimental results highlight an impressive offline training accuracy of 98.13% for mmMotion as well as a real-time motion recognition rate of 90.5%. The robustness of this system against various motions and its excellent generalization across different individuals further underscores its potential as a viable solution for smart home applications. Future work will focus on expanding mmMotion to support multiperson detection while enhancing its applicability in more complex environments to enable comprehensive monitoring within smart home IoT scenarios.

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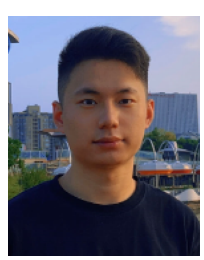

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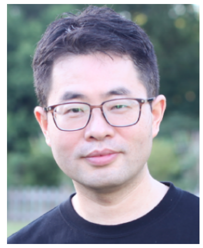

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